Machine learning

Generated on: 18/6/2026

Machine Learning Notes

Machine Learning (ML) is a branch of Artificial Intelligence that allows computers to learn patterns from data and make predictions or decisions without being explicitly programmed.

Core Concepts

- **Data**: Raw information used for training. Includes **Features** (inputs) and **Labels** (outputs).
- **Model**: Mathematical function that learns patterns from data.
- **Training**: Process of feeding data to the model so it adjusts its parameters to minimize errors.
- **Inference**: Using the trained model to make predictions on new data.
- **Dataset Split**:
  - Training Set (70-80%)
  - Validation Set (for tuning)
  - Test Set (final evaluation)

Types of Machine Learning

1. Supervised Learning
Model learns from labeled data.
- **Regression**: Predict continuous values (e.g., house price).
- **Classification**: Predict categories (e.g., spam/not spam).

2. Unsupervised Learning
Model finds patterns in unlabeled data.

- **Clustering**: Group similar data.
- **Dimensionality Reduction**.

3. Reinforcement Learning
Agent learns by trial and error using rewards.

4. Semi-Supervised Learning
Uses mostly unlabeled + some labeled data.

Key Concepts

- **Overfitting**: Model performs well on training data but poorly on new data.
- **Underfitting**: Model is too simple.
- **Bias-Variance Tradeoff**: Balance between underfitting and overfitting.
- **Regularization**: L1 (Lasso) and L2 (Ridge) to prevent overfitting.
- **Cross-Validation**: K-Fold CV for reliable evaluation.

Loss Functions

Loss functions measure how well the model’s predictions match the actual data. The goal of training is to **minimize** the loss.

Common Loss Functions

1. Mean Squared Error (MSE) – Used in Regression
$$
\text{MSE} = \frac{1}{n} \sum_{i=1}^{n} (y_i - \hat{y}_i)^2
$$

2. Mean Absolute Error (MAE) – Used in Regression

$$
\text{MAE} = \frac{1}{n} \sum_{i=1}^{n} |y_i - \hat{y}_i|
$$

3. Binary Cross-Entropy (Log Loss) – Used in Binary Classification

$$
\text{BCE} = -\frac{1}{n} \sum_{i=1}^{n} \left[y_i \log(\hat{y}_i) + (1 - y_i) \log(1 - \hat{y}_i) \right]
$$

4. Categorical Cross-Entropy – Used in Multi-class Classification

$$
\text{CCE} = -\sum_{i=1}^{n} \sum_{c=1}^{C} y_{i,c} \log(\hat{y}_{i,c})
$$

5. Hinge Loss – Used in Support Vector Machines (SVM)

$$
\text{Hinge} = \max(0, 1 - y_i \cdot \hat{y}_i)
$$

Evaluation Metrics

Task	Metrics
Regression	MSE, RMSE, MAE, R^2 Score
Classification	Accuracy, Precision, Recall, F1-Score, AUC-ROC
Clustering	Silhouette Score

Machine Learning Pipeline

1. Problem Definition
2. Data Collection & EDA
3. Data Preprocessing
4. Feature Engineering
5. Model Selection & Training
6. Hyperparameter Tuning
7. Evaluation
8. Deployment & Monitoring

Popular Algorithms

- **Classical**: Linear Regression, Decision Trees, Random Forest, SVM, KNN.
- **Ensemble**: XGBoost, LightGBM.
- **Deep Learning**: Neural Networks, CNNs, RNNs/LSTMs, Transformers.

Tools & Libraries

- **Python**: scikit-learn, pandas, NumPy, Matplotlib, TensorFlow, PyTorch.

Applications

- Recommendation Systems, Image Recognition, NLP, Fraud Detection, Healthcare, Autonomous Vehicles.

Challenges

- Data Quality, Overfitting, Interpretability, Bias & Fairness, Computational Cost.

Tip: Start with scikit-learn and Andrew Ng’s Machine Learning course.

Brief Revision Notes - Updated June 2026